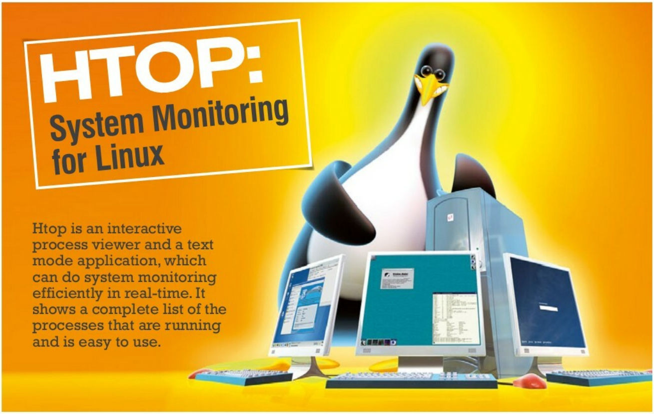
HTOP: System Monitoring for Linux



Htop is an interactive process viewer and a text mode application, which can do system monitoring efficiently in real-time. It shows a complete list of the processes that are running and is easy to use.

System monitoring is a very important activity for any organisation. A system should always be free from malfunctions and should be continuously available to serve its purpose, and for this to happen, efficient system monitoring is a must. One of the most important responsibilities a systems administrator has is to monitor the systems in the network. The systems administrator should be aware of what processes are running on the system, how much memory or CPU resources are being utilised, who is logged in and so on. Good system monitoring can help in maintaining operational continuity and avert any unwanted performance issues. The task of system monitoring can be efficiently done with various sets of tools. GNU/Linux provides various inbuilt utilities for this purpose. One such utility is the *top* command, which displays a real-time list of processes that are running on the system. It also displays additional information about the system uptime, current CPU and memory usage (or the total number of running processes), and allows you to perform actions such as sorting the list or killing a process. The screen constantly updates itself.

Htop, which is a *ncurses* based process viewer, offers some very interesting options. It supports mouse operation, uses colour in its output and gives visual indications about the processor, memory and swap usage. Htop also prints full command lines for processes, and allows one to scroll both vertically and horizontally for processes and command lines, respectively. In this article, let us explore what Htop has to offer.

To install Htop on Debian based Linux systems, use the command given below:

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$ sudo apt-get install htop
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Htop can be run from the command line by typing the command below:

```
$ htop
```

The output can be broken down into three parts—the header, body and footer.

The *header* displays CPU information—the numbers 1 and 2 (top-left in Figure 1) represent the number of cores in the system. It also displays memory usage, swap usage and loads the average on the right in the form of text/counters. The progress bars for CPU information are of different colours. The following list will explain what each colour means.

Blue: Low priority processes
Green: User processes
Red: Kernel processes
Yellow: IRQ (interrupt request) time
Magenta: Soft IRQ (interrupt request) time
Grey: IO wait time

Memory and swap bars are also in different colours. Here is a list of what the colours mean in relation to the memory and swap progress bars.

Green: Used memory pages
Blue: Buffer pages
Yellow: Cache pages

The *body* section lists all the running processes on a